

Scientists use "empty space" to strengthen superconductivity

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Scientists have shown that tiny changes called **quantum fluctuations** in empty space can strengthen superconductivity. The source report, published on October 3, 2026, calls this the first experiment to show the effect.

A **superconductor** is a material that carries electric current without electrical resistance under certain conditions. Its **critical temperature** is the temperature below which it becomes superconducting. Raising this temperature lets it work in slightly warmer conditions.

The study appeared in Nature. Changgan Zeng and Guanghui Cheng at the University of Science and Technology of China led it. Researchers from Shanghai Jiao Tong University, the Massachusetts Institute of Technology and other partners also helped.

The team studied a surprising idea: empty space, called a **vacuum**, is not completely still. Even at its lowest energy, it has tiny **quantum fluctuations**. Scientists have seen effects of these fluctuations in other experiments.

Zeng explained that these fluctuations in ordinary empty space are usually too weak to have visible effects on large systems of materials. The team used a special structure called a **terahertz dark cavity** to strengthen their influence. It changes the electromagnetic surroundings of a material.

They put a superconducting material called NbSe₂ inside the cavity. They compared how it worked inside and outside.

In a device with six layers of NbSe₂, the **critical temperature** rose by up to 5.4%. Cheng reported that the **critical current** and critical magnetic field also rose greatly near the point where superconductivity begins. These are limits on the current and magnetic field the material can handle while staying superconducting.

The researchers changed parts of the experiment to test other explanations. Their checks ruled out effects such as strain, material damage and certain effects from nearby metal.

The improvement was greatest at one particular cavity frequency. Zeng said this was strong evidence that the superconducting state was interacting with electromagnetic patterns in the cavity. The findings offer a new way to study and control unusual states of matter.

Word Treasure

superconductor -- A material that carries electric current without electrical resistance under certain conditions.

critical temperature -- The temperature below which a material becomes superconducting.

vacuum -- Empty space that still has tiny quantum fluctuations.

quantum fluctuations -- Tiny changes that happen even in a vacuum at its lowest energy.

terahertz dark cavity -- The special structure used to change a material's electromagnetic surroundings.

critical current -- A limit on the current a material can handle while staying

superconducting.

Break it down

The opening presents the finding and the source report's date.

The article explains superconductors and critical temperature, then names the study leaders and partners.

It explains vacuum fluctuations and describes the cavity experiment.

It reports the results, checks of other explanations and frequency evidence.

Why it matters

The experiment offers a new way to study and control superconductivity. A higher critical temperature allows the material to remain superconducting in slightly warmer conditions.

Test what you remember

1. What can a superconductor do under certain conditions?

- ☐ (A) Stop quantum fluctuations
- ☐ (B) Remove all magnetic fields
- ☐ (C) Turn metal into empty space
- ☐ (D) Carry electric current without electrical resistance

2. Who led the study?

- ☐ (A) Changgan Zeng and Guanghui Cheng
- ☐ (B) All the other partners
- ☐ (C) Only researchers from Shanghai Jiao Tong University
- ☐ (D) Only researchers from the Massachusetts Institute of Technology

3. What material did the researchers put inside the cavity?

- ☐ (A) A wooden block
- ☐ (B) NbSe₂
- ☐ (C) A piece of glass
- ☐ (D) A plastic sheet

4. How much did the critical temperature rise in the six-layer device?

- ☐ (A) Up to 5.4%
- ☐ (B) It did not rise
- ☐ (C) Exactly 5.4 degrees
- ☐ (D) Up to 54%

5. Why did the team change parts of the experiment?

- ☐ (A) To stop the material carrying current
- ☐ (B) To remove all six layers
- ☐ (C) To test other explanations for the results
- ☐ (D) To change the study's publication date

6. When was the improvement greatest?

- ☐ (A) Only after the material was damaged
- ☐ (B) At one particular cavity frequency
- ☐ (C) At all cavity frequencies equally
- ☐ (D) Outside the cavity at every frequency

Answer key (try the quiz first!): 1: D 2: A 3: B 4: A 5: C 6: B

1. **Changgan Zeng:** Zeng explained why the team strengthened the influence of vacuum fluctuations and said the frequency result provided strong evidence of interaction with the cavity.
2. **Guanghai Cheng:** Cheng reported that the critical current and critical magnetic field also rose greatly near the point where superconductivity begins.